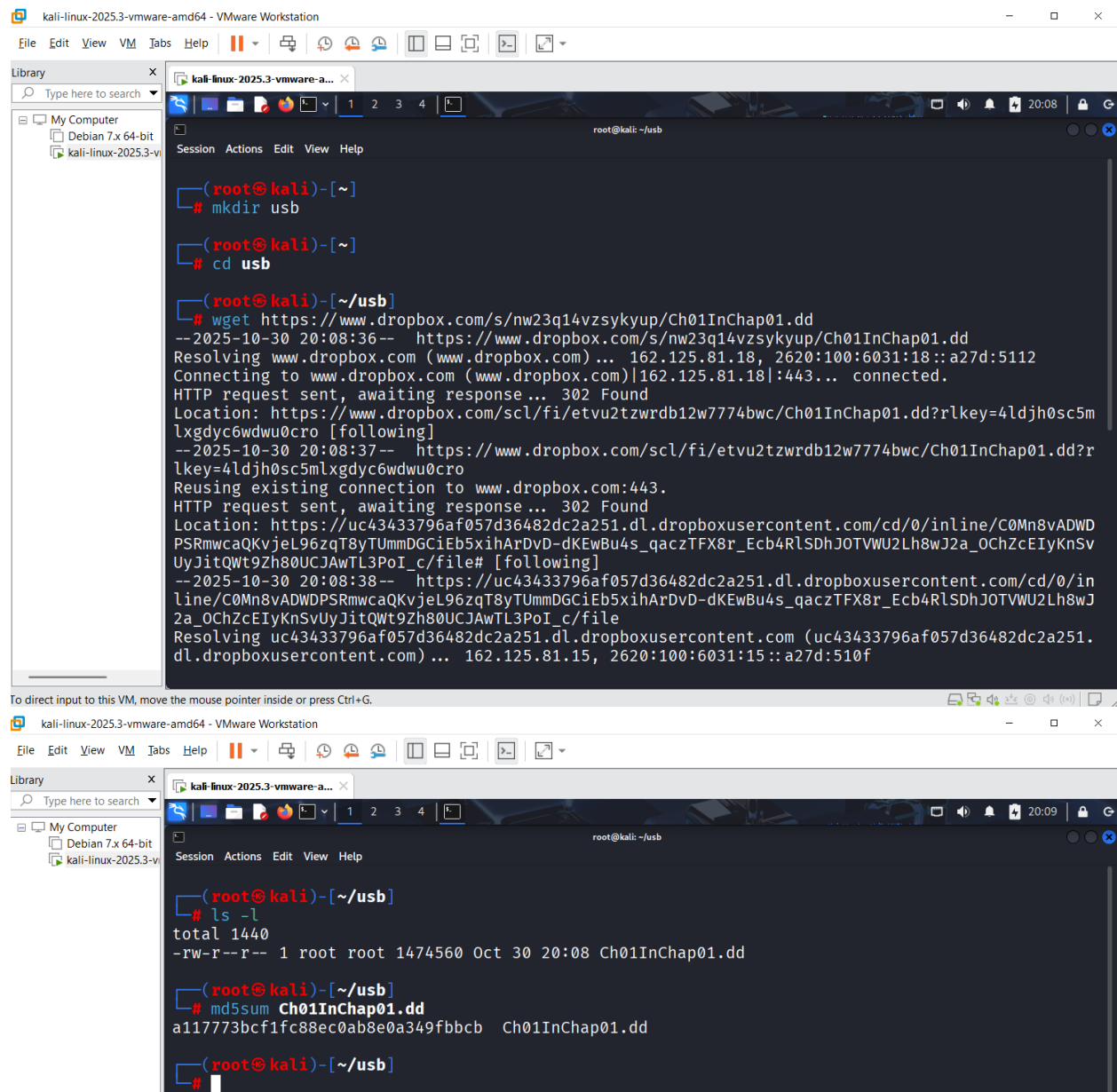
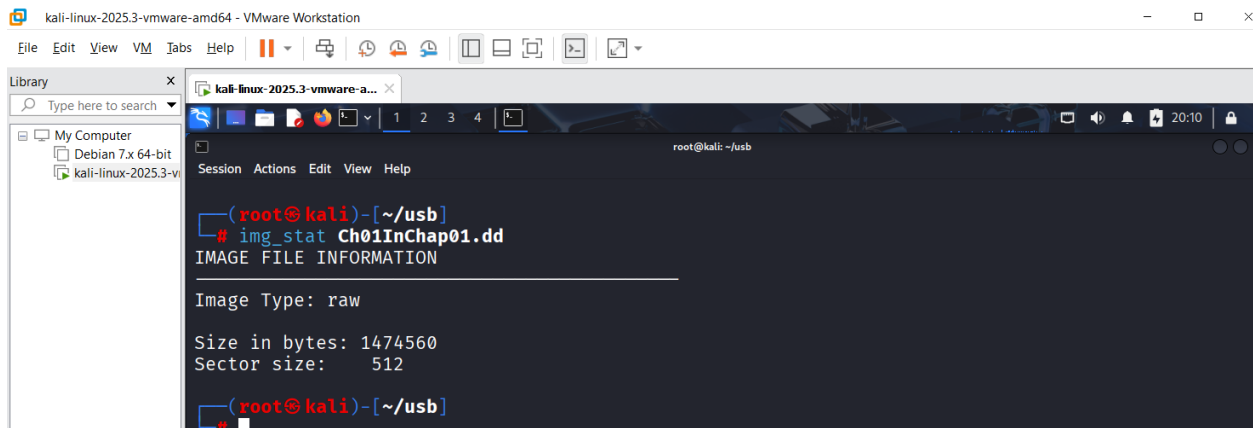


LAB 13



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vm
kali-linux-2025.3-vmware-a...
root@kali: ~usb
Session Actions Edit View Help
(root@kali)-[~]
# mkdir usb
(root@kali)-[~]
# cd usb
(root@kali)-[~/usb]
# wget https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
--2025-10-30 20:08:36-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
Resolving www.dropbox.com (www.dropbox.com)... 162.125.81.18, 2620:100:6031:18::a27d:5112
Connecting to www.dropbox.com (www.dropbox.com)|162.125.81.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5m
lxgdy6wdwu0cro [following]
--2025-10-30 20:08:37-- https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?r
lkey=4ldjh0sc5mLxgdy6wdwu0cro
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc43433796af057d36482dc2a251.dl.dropboxusercontent.com/cd/0/inline/C0Mn8vADWD
PSRmwcaQKvjel96zqT8yTUmmdGCIeb5xihArDvD-dKEwBu4s_qaczTFX8r_Ecb4RLSDhJOTVWU2Lh8wJ2a_0ChZcEIyKnSv
UyJitQWt9Zh80UCJAwtL3PoI_c/file# [following]
--2025-10-30 20:08:38-- https://uc43433796af057d36482dc2a251.dl.dropboxusercontent.com/cd/0/in
line/C0Mn8vADWDPSRmwcaQKvjel96zqT8yTUmmdGCIeb5xihArDvD-dKEwBu4s_qaczTFX8r_Ecb4RLSDhJOTVWU2Lh8wJ
2a_0ChZcEIyKnSvUyJitQWt9Zh80UCJAwtL3PoI_c/file
Resolving uc43433796af057d36482dc2a251.dl.dropboxusercontent.com (uc43433796af057d36482dc2a251.
dl.dropboxusercontent.com)... 162.125.81.15, 2620:100:6031:15::a27d:510f
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vm
kali-linux-2025.3-vmware-a...
root@kali: ~usb
Session Actions Edit View Help
(root@kali)-[~/usb]
# ls -l
total 1440
-rw-r--r-- 1 root root 1474560 Oct 30 20:08 Ch01InChap01.dd
(root@kali)-[~/usb]
# md5sum Ch01InChap01.dd
a11773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
(root@kali)-[~/usb]
#
```

2 ảnh trên: chuẩn bị file và tải ảnh đĩa cho bài lab



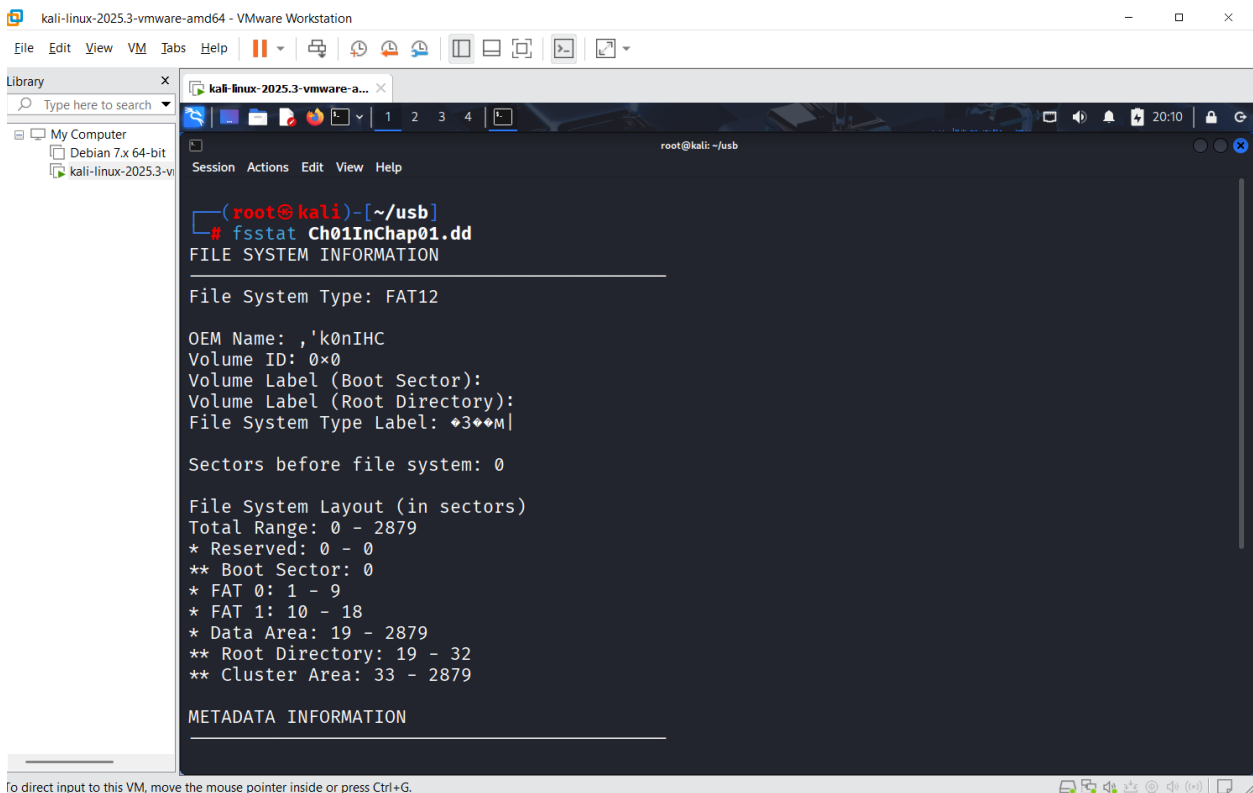
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vm
kali-linux-2025.3-vmware-a...
root@kali: ~/usb
Session Actions Edit View Help
(root@kali)-[~/usb]
# img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
Image Type: raw
Size in bytes: 1474560
Sector size: 512
(root@kali)-[~/usb]
#
```

ảnh trên: Thông tin ảnh đĩa sử dụng lệnh `img_stat` ta thấy

Image Type: raw

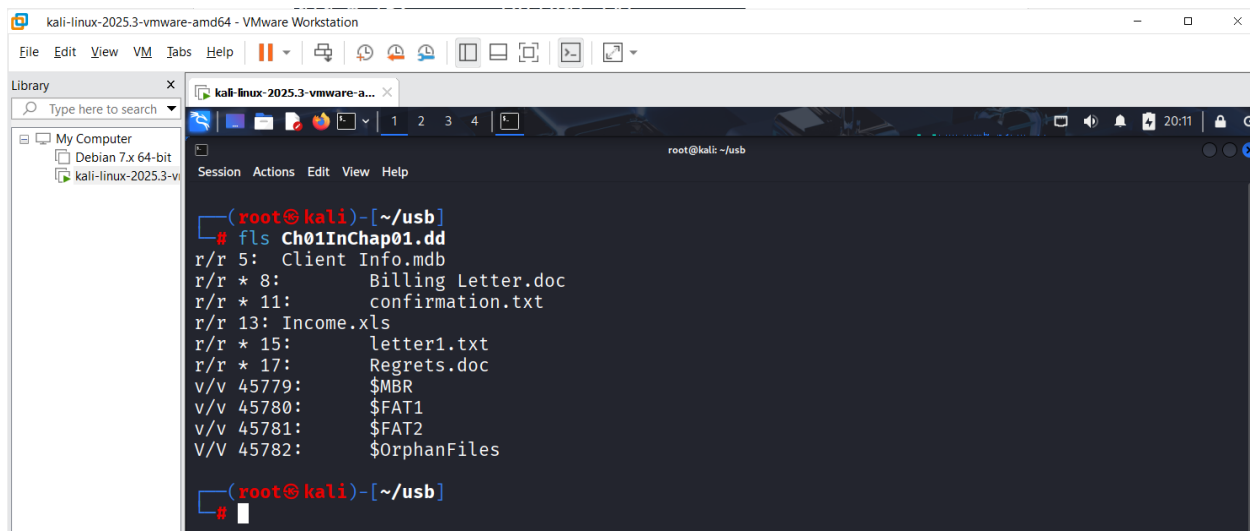
Size in bytes: 1474560

Sector size: 5



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-vm
kali-linux-2025.3-vmware-a...
root@kali: ~/usb
Session Actions Edit View Help
(root@kali)-[~/usb]
# fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: +3+M
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
METADATA INFORMATION
```

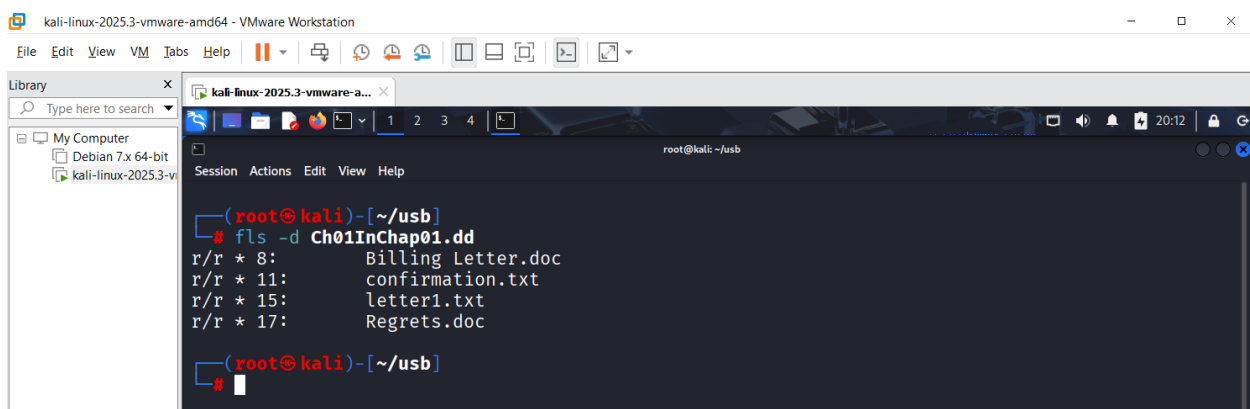
ảnh trên: xác định loại FS (FAT/NTFS/ext...), kích thước cluster, label, root dir, v.v. bằng lệnh `fsstat`



```
(root@kali)~/usb
# ls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(root@kali)~/usb
#
```

ảnh trên: dùng lệnh `ls` kết quả sẽ hiển thị các tệp đã cấp phát (r/r) và các tệp đã xóa (r/r *)

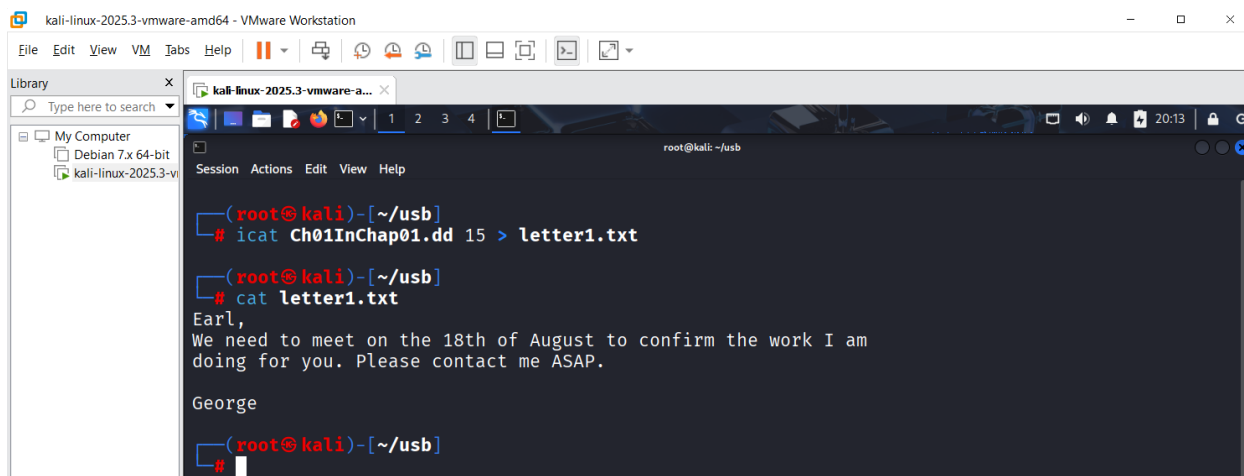


```
(root@kali)~/usb
# ls -d Ch01InChap01.dd
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc

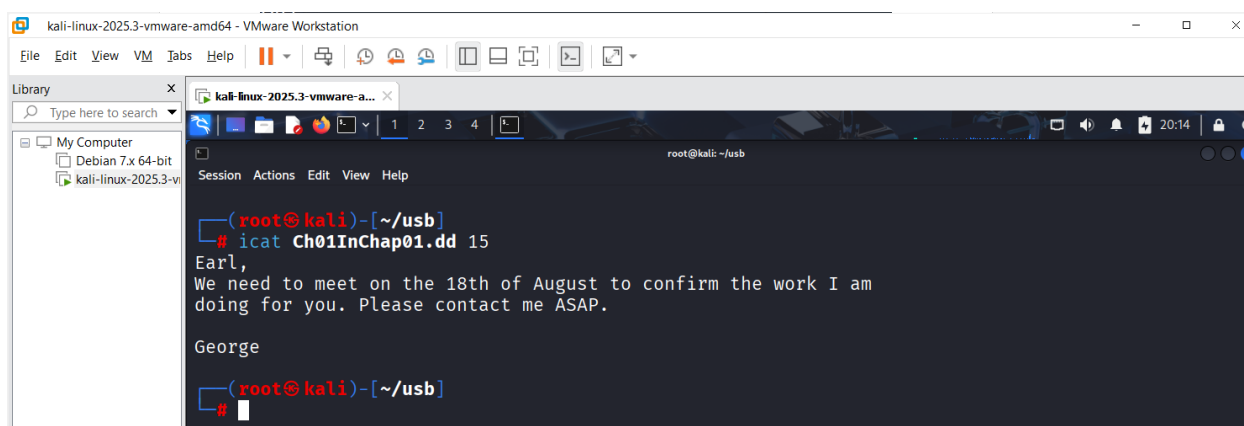
(root@kali)~/usb
#
```

ảnh trên: dùng lệnh `ls` nhưng thêm tùy chọn `-d` để chỉ xem những file đã bị xóa và xem được inode của từng file

Có 4 tệp bị xóa: Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc.



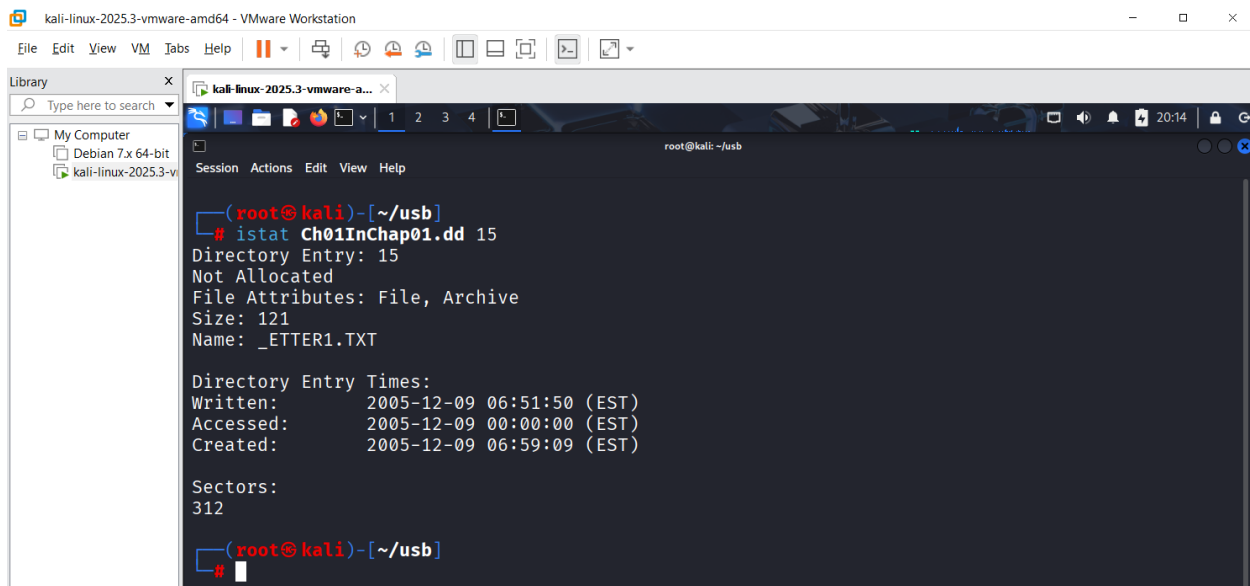
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-v...
kali-linux-2025.3-vmware-a...
root@kali: ~/usb
Session Actions Edit View Help
(root@kali)~/usb
# icat Ch01InChap01.dd 15 > letter1.txt
(root@kali)~/usb
# cat letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.
George
(root@kali)~/usb
#
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Debian 7.x 64-bit
kali-linux-2025.3-v...
kali-linux-2025.3-vmware-a...
root@kali: ~/usb
Session Actions Edit View Help
(root@kali)~/usb
# icat Ch01InChap01.dd 15
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.
George
(root@kali)~/usb
#
```

2 ảnh trên: dùng lệnh `icat` để khôi phục file đã bị xóa (`letter1.txt`) dựa vào inode (xem được bằng lệnh `fls` ở trên ta thấy inode của file `letter1.txt` là 15) và xem nội dung tệp.

Lệnh `icat` xuất **nội dung file** theo inode, không cần mount.



```
(root@kali)~[~/usb]
# istat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:09 (EST)

Sectors:
312

(root@kali)~[~/usb]
#
```

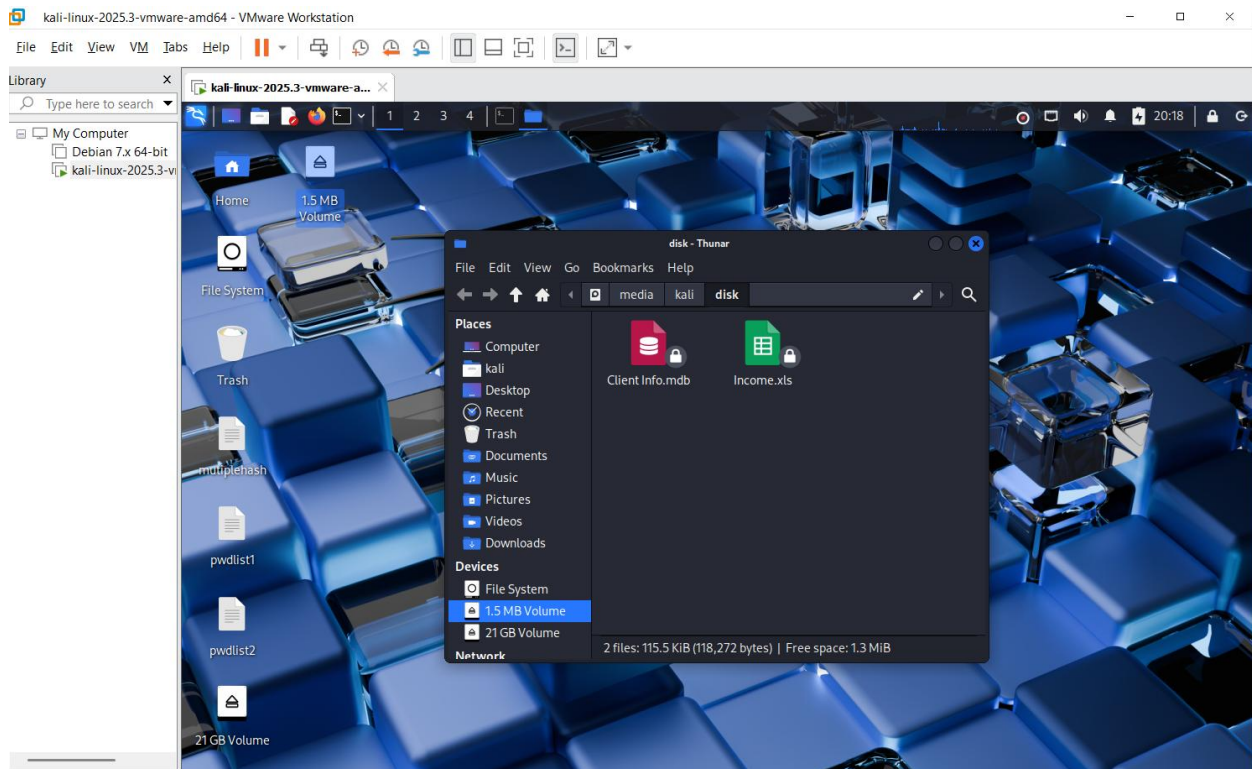
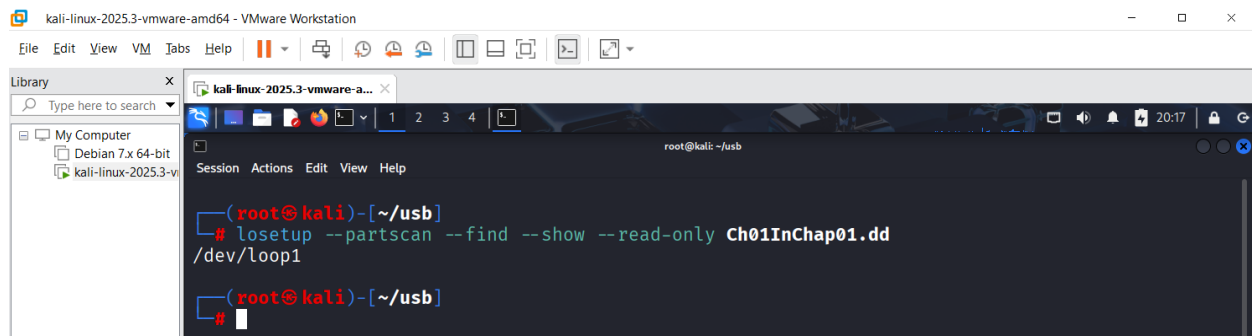
ảnh trên: dùng **istat** xem **metadata của một file/đối tượng** theo inode, từ đó biết file có còn allocate không, kích thước, thời gian MAC, và **các block/cluster** mà file chiếm dụng

Kết quả sẽ hiển thị: Not Allocated (đã bị xóa), Size: 121, và Sectors: 312

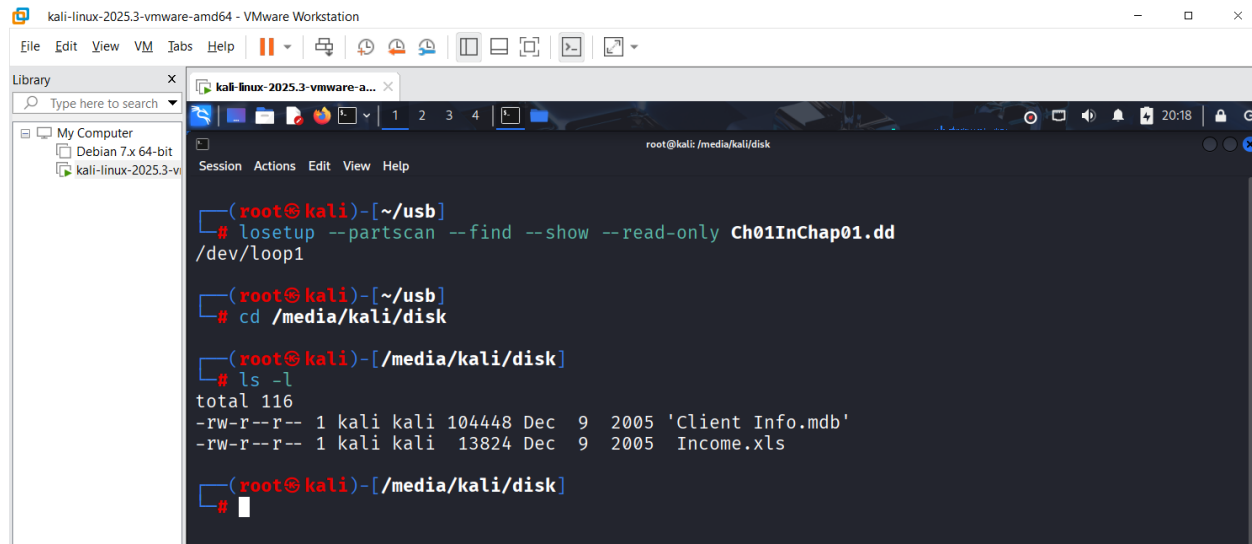
The first screenshot shows a terminal session where the command `blkls Ch01InChap01.dd 312-312` is executed. The output is a message from George to Earl: "Earl, We need to meet on the 18th of August to confirm the work I am doing for you. Please contact me ASAP. George".

The second screenshot shows the same terminal session where the command `blkcat Ch01InChap01.dd 312 | xxd` is executed. The output displays the raw hex data of the message, with the text "Earl, ..We need to meet on the 18th of August to confirm the work I am ..doing for you. Please contact me ASAP.. ... George....." visible on the right side of the hex dump.

ảnh trên: dùng lệnh blkls và cat để thao tác với sector Kết quả sẽ hiển thị nội dung sector 312 ở dạng Hex và văn bản



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



3 ảnh trên: mount phân vùng ảnh đĩa và truy cập, ta thấy dù không cần mount phân vùng vẫn có thể dùng các lệnh để thao tác với ảnh đĩa